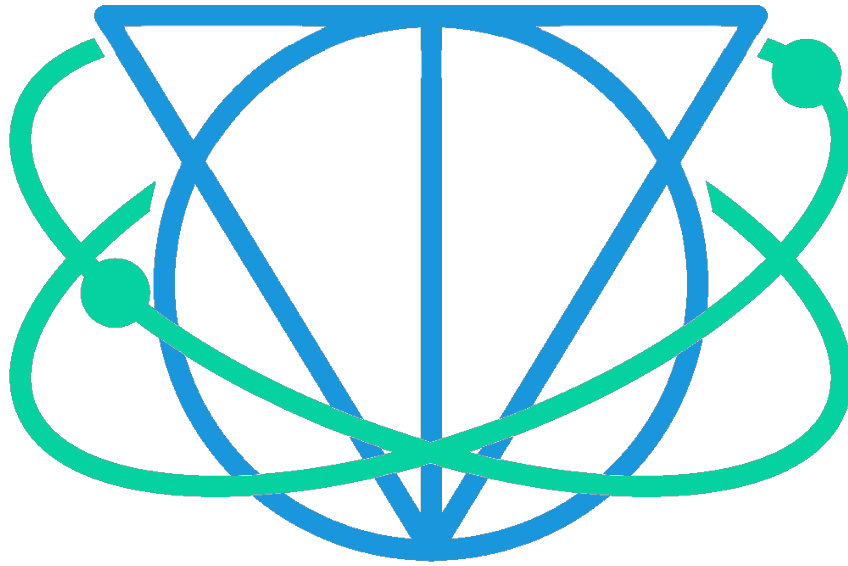




2026 Thomas Jefferson Physics Olympiad

We acknowledge the following people for their contributions to this year's exam (in alphabetical order):

Aarush Deshpande, Eric Xie, Kabeer Parmar, Noah Kops, Tiger Deng

Instructions

This year's contest lasts from **June 27, 2026 12A.M. EST** to **June 30, 2026 12 A.M. EST**.

- Please join the discord for the latest exam updates: <https://discord.gg/gATxvtx9R>.
- All clarifications should be emailed to tjhsstphysics team@gmail.com
- Use $g = 9.8 \frac{\text{m}}{\text{s}^2}$ **unless otherwise specified**.
- Questions are not necessarily in order of difficulty, so try all of them!
- The final solution should be a write-up, clearly detailing all the steps to reach a final solution, followed by a **boxed** final answer.
- Handwriting counts - if it is illegible, it is not being graded. If you are unsure if your handwriting is legible, typeset your work with LaTeX or Typst. Keep in mind that cursive is also hard to read.
- Answers or steps that are not dimensionally correct shall receive 0 points.
- Working that is difficult to follow shall receive 0 points.
- You are welcome to use Computer Algebra Systems, including (but not limited to) SymPy, Wolfram, and Matlab.
- Generative AI is **strictly forbidden**, and any usage will result in disqualification. This competition is just for fun, please don't ruin it.
- **Do not communicate information to anyone except your team members before the end of the competition**

Notation Review

- The symbol c will be used to denote the speed of light, unless otherwise stated.
- An average of a quantity v is denoted as follows: $\langle v \rangle$
- The notation ∂F denotes the boundary of a manifold F . For example, if S is a surface then ∂S is the curve that is the boundary of the surface. Likewise, if V is a volume, then ∂V is the surface of the volume.
- For flux integrals, we may use $d\mathbf{S} \equiv \mathbf{n} dA$, where $\mathbf{n}$ is the assumed well defined normal vector at each point.
- For volume and surface integrals, we will simply use one integral, i.e. $\int_V F dV$ instead of $\iiint_V F dV$.

Mathematical Prerequisites

Problems marked with a [A] may require some of the following theorems from multivariable calculus:

$$\begin{aligned}\int_a^b \nabla f \cdot d\mathbf{s} &= f(b) - f(a) \\ \int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} &= \int_{\partial S} \mathbf{F} \cdot d\vec{\ell} \\ \int_V \nabla \cdot \mathbf{F} dV &= \int_{\partial V} \mathbf{F} \cdot d\mathbf{S}\end{aligned}$$

where ∇ is the “vector” of partial derivatives in Cartesian coordinates $\langle \partial_x, \partial_y, \partial_z \rangle$, $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ is a scalar field, and $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a vector field. In general, these problems will be worth less points, and are only included to demonstrate some interesting phenomena.

Problems

1. Marangoni Stresses

In the 1860s, Carlo Marangoni showed that surface tension gradients can actually produce shear forces that we call Marangoni stresses. Such surface tension gradients arise from either gradients in temperature, or changes in the chemical composition at the interface.

- A) [A] Show that a fluid configuration cannot be static if such surface tension gradients exist
- B) Consider a spherical soap bubble of radius R , mass density ρ , and thickness $\tau \ll R$ in a vacuum. An external agent then slowly places the bubble in a uniform gravitational field $-g_0 \hat{z}$. Assuming the bubble's radius stays approximately constant, what is the change in the surface energy of the bubble?

2. Spinny Spinny Dipole

Consider a classical model of the electron (a charge carrier with net charge $-e$) as a rotating magnetic dipole with magnetic dipole moment $p_m = e\hbar/2m$. If the dipole's angular velocity is ω , find $\hbar\omega$ in terms of the rest energy of the electron, $E = mc^2$. You should treat the problem classically, i.e. $\gamma \approx 1$ everywhere in space, where γ is the Lorentz factor.

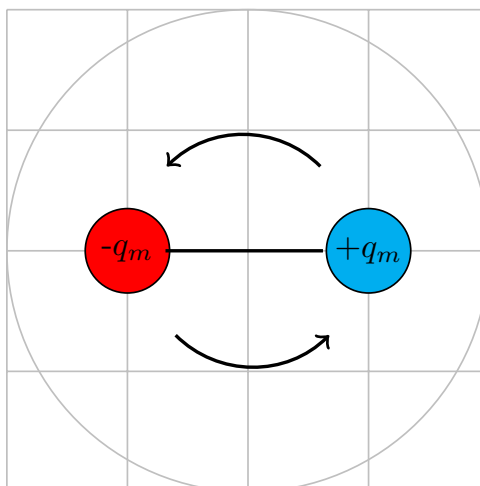


Figure 1: The electron, modeled as a magnetic dipole rotating in the plane of the figure

3. A Dangerous Atmosphere

The atmosphere of a planet consists of Uranium Hexafluoride (UF_6) gas. What is the average height of the gas in the atmosphere, relative to the surface of the planet? You may assume that $g = 9.81 \frac{\text{m}}{\text{s}^2}$ and the temperature $T = 80^\circ \text{C}$ are both approximately constant across the atmosphere, and that UF_6 gas behaves ideally.

4. Spin the Bottle

A cylinder of radius r is filled to a height h with a liquid with density ρ . The coefficient of friction between the liquid and the cylinder is μ . The cylinder is pivoted at the bottom end (Figure 2), and begins to rotate with a constant angular velocity ω .

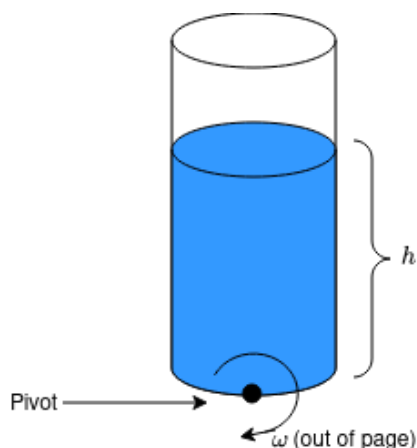
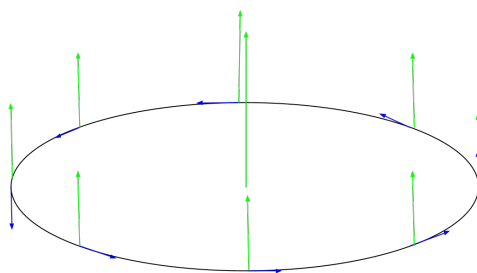


Figure 2: The cylinder is pivoted, and rotates with ω pointing perpendicular to the page

Let the surface tensions of the liquid/air, liquid/cylinder, and cylinder/air interfaces be γ_{LA} , γ_{LC} , and γ_{CA} , respectively. What is the maximum $\omega =: \omega_c$ such that the fluid stays inside the cylinder?

5. Particle Accelerator



Eric wants to construct a simple, circular particle accelerator of radius r to accelerate a particle of charge q and mass m . He plans on generating a time-varying magnetic field within the particle accelerator of average magnitude $\langle B_{\text{within}} \rangle(t) = kt$, using the resulting electric field to speed up the particle and the resulting magnetic field to turn it in an orbit. Assume that the magnetic field is always parallel to the axis of symmetry of the particle accelerator and that the electric field is always pointing parallel to the velocity of the particle. In order for the magnetic field to simultaneously curve the particle and accelerate it correctly, the following condition is required as a consequence of Faraday's Law of Induction: $\Delta B_{\text{loop}} = \frac{1}{2} \Delta \langle B_{\text{within}} \rangle$. Calculate the time derivative of the speed v of the particle as a function of v . Assume $v \sim c$.

6. Extreme Weather

In late August 2005, Hurricane Katrina struck Louisiana. Reports show that near its landfall, the hurricane had maximum wind speeds of around 120 mph (54 m/s) and a radius of maximum wind speed of about 30 miles (48 km). Here, we will model Katrina's inner vortex core using a Rankine vortex model.

Given that maximum tangential wind speed $v_{\theta, \text{max}} = 120$ mph and radius of maximum wind speed $R_c = 30$ miles, and assuming that the maximum tangential wind speed occurs at the outer edge of the vortex core ($r = R_c$)

Derive:

- A) An expression and value for the vertical vorticity inside the vortex core
- B) An expression and value for the time required for an air parcel inside the vortex core to complete one full revolution

7. Suboptimal Induction Heating

Suppose Gyrados constructs a hollow, thin-walled cylinder of radius R and height H without top and bottom faces using a single rectangular sheet of conductive material of thickness $\varepsilon \ll R$, resistivity ρ , mass m , and heat capacity c_p . If Gyrados places this cylinder in a uniform, time-varying magnetic field of magnitude $B = A \sin(\omega t)$ which is parallel to the axis of symmetry of the cylinder, it will induce a current in the cylinder which will produce heat according to $P = IV$. However, in real life, heating this cylinder would cause it to expand and cause the resistivity to change. The following equations govern how the change in temperature affects the properties of the cylinder: $H = H_0 + \alpha \Delta T$, $R = R_0 + \alpha \frac{R_0}{H_0} \Delta T$, $\rho = \rho_0 + \alpha_1 \Delta T$, and $\varepsilon = \varepsilon_0 + \alpha \frac{\varepsilon_0}{H_0} \Delta T$.

- A) Given initial temperature T_0 , create a differential equation isolating $\frac{dT}{dt}$ on one side, defining it in terms of R_0 , H_0 , ρ_0 , ε_0 , α , α_1 , m , c_p , T , T_0 , A , ω , and t .
- B) Sketch a graph of the solution to this differential equation with labels.

Consider changes in the magnetic field from the induced current negligible.

8. Statistical Physics and Neural Networks

- A) A one-dimensional Ising paramagnet at temperature T consists of N spins in a line. The spins only interact with an external magnetic field and not with each other. The system state is described by a column vector $\vec{\sigma} = [\sigma_1 \cdots \sigma_N]^T$ where σ_i are spin variables (they are ± 1). Its Hamiltonian¹ is

$$\mathcal{H}(\vec{\sigma}) = -\mu_B B \sum_{i=1}^N \sigma_i$$

where B is an external magnetic field and μ_B is the Bohr magneton.

- i) Find the equation of state for $m := \langle \sigma \rangle$, the magnetization per spin.
- ii) Now, we allow the spins to interact with each other by replacing $\mu_B B$ by an effective field, $\mu_B B_e = \mu_B B + Jm$ where $J > 0$ is a coupling between the spins and m is the magnetization. By adding a coupling between spins, we introduce a ferromagnetic state. Find the new equation of state for the magnetization per spin.
- iii) Find the critical temperature T_c for which this system transitions from a paramagnet to a ferromagnet.
- B) There is no requirement for the coupling J to be homogeneous between every pair of spins. In fact, this is not true for many materials. Therefore, we introduce the spin glass Hamiltonian,

$$\mathcal{H}(\vec{\sigma}) = -\frac{1}{2N} \sum_{i=1}^N \sum_{j=1}^N J_{ij} \sigma_i \sigma_j - \mu_B B \sum_{i=1}^N \sigma_i$$

where $J_{ij} = J_{ji}$ and is distributed normally about 0

- i) Find the local effective magnetic field experienced by a particular spin.

¹The total energy

- ii) Find the change in energy ΔE after flipping a given spin σ_i
- iii) Show that the energy decreases monotonically as dynamics continue. Explain why the final configuration is a “frozen” state.
- iv) To characterize the onset of the “freezing,” we introduce an order parameter that measures whether spins remain frozen in time.

$$q = \frac{1}{N} \sum_{i=1}^N \langle \sigma_i \rangle^2$$

q also satisfies the equation,

$$q = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{z^2}{2}\right) \tanh^2(\beta J \sqrt{q} z) dz$$

where $\beta := \frac{1}{k_B T}$. Find the critical temperature T_c below which a non-zero solution for q appears.

- C) We can also ask whether memory can emerge from the collective interactions of simple units. From the parts above, you have shown how the observable macroscopic state of a system is related to the coupling between spins. You have also shown that when these couplings vary between pairs, the system can encode “frozen” states. If we can encode frozen states in the structure of the Hamiltonian $\mathcal{H}$ itself, the system will naturally flow towards these frozen states, or memories. A memory $\vec{\xi}^{(\mu)}$ is indexed by μ out of a total of P stored memories. Now, we construct a network of N “neurons,” and map it to our spin network. If a particular neuron fires, it has $\sigma_i = 1$. Similarly, if it is silent, it has spin $\sigma_i = -1$. In this network, a memory $\vec{\xi}^{(\mu)}$ is just a subset of neurons firing together.
- i) Create a function of the overlap between a network microstate n with the memory μ , $M(\vec{\xi}^{(\mu)}, \vec{\sigma}^{(n)})$
 - ii) When $M(\vec{\xi}^{(\mu)}, \vec{\sigma}^{(n)}) = \pm 1$, the state of the network exactly matches that of the memory (or antimemory), $\vec{\sigma}^{(n)} = \pm \vec{\xi}^{(\mu)}$. Use your overlap function $M(\vec{\xi}^{(\mu)}, \vec{\sigma}^{(n)})$, and the internal energy of the ferromagnet $U = \langle \mathcal{H} \rangle = -\frac{1}{2} J N m^2 - \mu_B B N m$ to construct a Hamiltonian $\mathcal{H}$ for this system in terms of σ and an interaction weight,

$$W_{ij} = \frac{1}{N} \sum_{\mu=1}^P \xi_i^{(\mu)} \xi_j^{(\mu)}$$

- D) Now consider the dynamics of such a system. The network evolves in discrete time steps,

$$\vec{\sigma}(t) \rightarrow \vec{\sigma}(t+1)$$

The Hamiltonian also becomes time dependent,

$$\mathcal{H}(t) = -\frac{1}{2} \sum_{i=1}^N h_i(t) \sigma_i(t)$$

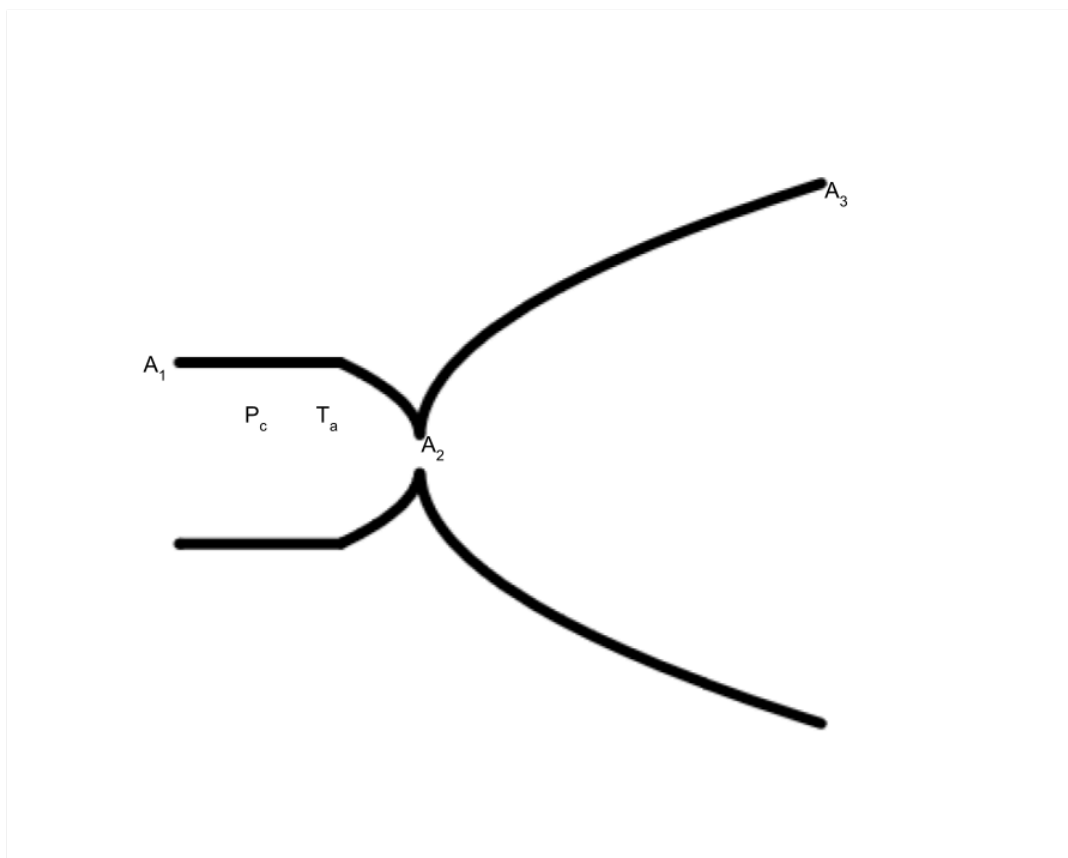
where $h_i(t)$ is the effective field acting on a neuron i ,

$$h_i(t) = \sum_{j=1}^N W_{ij} \sigma_j(t) + \mu_B B_i$$

Now, consider updating a single neuron k . Find the change in the Hamiltonian $\Delta \mathcal{H}(t)$ after one time step in terms of $h_k(t)$ and $\sigma_k(t)$.

- E) A network like this cannot store an infinite number of memories without making mistakes.
- Use your formula for $\sigma_i(t+1)$ from part D to determine the condition under which a memory can be properly recalled by the network.
 - Assuming the patterns are random, and that $N > P \gg 1$, find an approximate formula for the probability that a given memory in P will be recalled incorrectly. Give your answer in terms of N and P .

9. Ethanol Rocket



Bob is thirsty but only has access to 1000kg of 100% pure ethanol ($\text{C}_2\text{H}_5\text{OH}$) (46.07 g/mol), 2083.78554374kg pure oxygen (32.00 g/mol), and a functioning rocket of dry mass m_0 . He decides to combust the ethanol with the oxygen to produce water according to the following balanced equation: $\text{C}_2\text{H}_5\text{OH} + 3 \text{O}_2 \longrightarrow 3 \text{H}_2\text{O} + 2 \text{CO}_2$ which releases $1366.8 \frac{\text{kJ}}{\text{mol } \text{C}_2\text{H}_5\text{OH}}$. One mole is equal to $6.022 \cdot 10^{23} = N_A$ particles. His rocket releases the reactants into the reaction chamber at 10 moles per second and 30 moles per second respectively. Given chamber pressure P_c , temperature T_a , no heat loss, and specific heat capacities c_{p1} and c_{p2} for H_2O and CO_2 respectively, solve the following problems.

- Find the initial velocity v_0 of the products within the reaction chamber of area A_1 .
- Bob measures the temperature of the exhaust exiting the nozzle as T_e . Find the final exhaust velocity v_e of the products as they exit the right end of the nozzle of area A_3 after passing through the throat with area A_2 where they reached the speed of sound.
- Bob happens to be in a vacuum with absolutely no gravity and decides to let the rocket burn all of its fuel starting at rest. Calculate the total efficiency of this rocket's engine over the course of its entire burn (the portion of the fuel's total initial energy transferred to the rocket's kinetic energy).

- D) If Bob starts the rocket at a greater initial speed, it will gain more kinetic energy. He can keep increasing this initial speed until rocket's gain in kinetic energy is greater than the fuel's chemical potential energy. Explain how conservation of energy is not violated. Include a verbal explanation followed by a separate mathematical explanation.

10. Protein Shake

Jayden Daniels² was making a protein shake, which included protein powder molecules of radius r and mass m . Interestingly, he noticed when the protein powder was soggy, it stuck to the side of a mixer with radius $R \gg r$. Consider the mixer to contain a fluid with coefficient of viscosity η rotating everywhere at a constant angular velocity ω . Assume there are $N \gg 1$ protein powder molecules stuck to the side of the mixer and the fluid has a surface tension γ and is filled to a height h . Estimate (excluding dimensionless coefficients) the minimum coefficient of friction μ necessary for this phenomena to occur.

11. Push Harder

A long cart moves with a speed $v \sim c$. Sand is dropped in cart at a rate $\frac{dm}{dt} = \sigma$ in the ground frame, and the sand is always dropped at the same place over the ground. Alice is stationary behind the cart, and pushes the back of the cart³ to keep it moving at a constant speed v . Let F be the force between Alice's feet and the ground.

- A) Compute F in the ground frame.
 B) By the longitudinal force transformation of special relativity, F must be the same for an observer in the cart's frame. Verify this explicitly by computing F in the cart's frame.

You may assume the cart is made of a maximally rigid material.

12. Windy Windmills

A wind turbine can only extract a certain fraction of the kinetic energy of the air - if it were to extract all, then the air would stop and no wind would go through the turbine - causing it to extract no energy. This fraction has a theoretical maximum - evaluate it (and show work). Assume air is incompressible.

13. Spaceships Galore

Two spaceships start to accelerate with a constant proper acceleration $a_1 = 15 \frac{m}{s^2}$ and $a_2 = 30 \frac{m}{s^2}$ respectively. Both spaceships motions are in a straight line with respect to their starting planet. However, in the planet's frame, the second spaceship's direction is 10° from the first spaceships. In the first spaceship's frame, after a year, how far is the second spaceship?

²Greatest Quarterback in NFL history by the way

³You can assume the mass of Alice's arms are negligible